

SOFTWARE PROJECT SYNOPSIS

Community Driven Collaborative Workspace

Team Members

Rishit Joshi - IU2341230659

Manish - IU2341230683

Tathya Dubey - IU2041230187

Guide / Mentor

Prof. Rupal Kharya

Department

IITE - B.Tech - CSE

1. Introduction

This synopsis presents the Collaborative Community Whiteboard Platform, a web-based application that integrates an infinite drawing canvas, embedded document/spreadsheet/slide blocks, real-time multi-user collaboration, and a public/private community sharing system. It has been prepared for the project guide and evaluation committee as an overview of the proposed system ahead of detailed design and development.

In today's digital work and study environment, individuals and teams are forced to rely on a combination of separate applications to complete a single task — one tool for drawing or brainstorming, another for writing documents, another for spreadsheets or presentations, and yet another for communicating and sharing their work with others. This constant switching between disconnected platforms leads to loss of productivity, duplication of effort, and a fragmented overall experience.

The purpose of this project is to design and develop a Collaborative Community Whiteboard Platform — a single, unified, web-based workspace that combines the functionality of an infinite drawing canvas, simplified document/spreadsheet/slide creation tools, real-time multi-user collaboration, and a community-driven sharing system, all within one application. This synopsis introduces the motivation, scope, intended functionality, and expected outcomes of the proposed system, and serves as the foundation for the detailed system design and development that follows.

2. Problem Statement

Teams commonly rely on Miro or Excalidraw for brainstorming, Google Workspace or Microsoft Office for documents and spreadsheets, and community platforms such as Reddit or Discord for sharing ideas and gathering feedback. Because these tools are disconnected from one another, moving between them disrupts workflow, reduces productivity, and makes collaboration less efficient, as content and context must be carried over manually from one application to the next.

As a result, a student working on a group assignment, a professional preparing for a team meeting, or an artist wanting feedback on their sketches must all rely on multiple disconnected applications, losing collaborative context along the way. This gap between creation, structured productivity, and community sharing forms the core motivation for this project.

3. Literature Review / Existing System

Existing collaboration tools each solve only one part of the problem. Whiteboarding tools such as Miro, Excalidraw, and Microsoft Whiteboard offer excellent freeform canvases for sketching and diagramming but lack proper document, spreadsheet, or presentation capabilities. Office suites such as Microsoft Office and Google Workspace offer powerful document creation but provide no true infinite canvas or freehand drawing experience. Community-based platforms such as Reddit allow content sharing and discussion but offer no creation tools whatsoever.

4. Objectives

The primary objective is to build a browser-based platform where users can draw, write, organize, collaborate in real time, and share their work with a community — all from a single canvas-based workspace — without needing to switch between multiple unrelated applications. More specifically, the main goals of this project are to:

- Design and develop a real-time collaborative whiteboard.
- Embed simplified document, spreadsheet, and slide blocks within the same canvas.
- Enable public and private community sharing of boards.
- Provide version history and export support.

The platform is intended to serve a range of target users, namely students working on notes and exam preparation, teachers conducting interactive lessons, professionals running meetings and preparing reports, project managers organizing project work, and artists and designers seeking feedback on their sketches.

5. Scope

The system will allow individual and team users to create infinite canvas-based “Boards” that can contain freehand drawings, shapes, sticky notes, images, and simplified document, spreadsheet, and slide-style content blocks. Multiple users will be able to edit the same board simultaneously in real time, and boards can be saved privately or published to public or private communities, where other users can view, comment on, like, bookmark, and duplicate them.

The platform is designed for three broad use cases: professional collaboration (meetings, planning, reports), educational collaboration (classroom teaching, notes, group assignments), and creative collaboration (sketching, mind maps, digital art). The project is scoped realistically for completion by a student team within one academic year, and therefore does not attempt to fully replicate enterprise-grade office software; instead, it focuses on a functional, integrated Minimum Viable Product (MVP) with clearly identified future enhancements.

The following key features are considered in scope for the current MVP:

- An infinite canvas with drawing tools
- Real-time collaboration between multiple users
- Basic document, spreadsheet, and slide blocks
- Community publishing, with support for likes and comments
- Export of boards to PDF and PNG formats

The following features are considered out of scope for the current MVP and are instead planned as future enhancements:

- An AI-based whiteboard assistant
- A mobile application with offline mode
- Handwriting and OCR recognition
- Community events

6. Proposed System

The proposed system is an infinite canvas workspace that allows users to add drawings, documents, spreadsheets, and slides as panels on the same board, supports real-time multi-user editing, and allows boards to be shared publicly or privately within a community.

This design gives the proposed system a clear advantage over existing tools: unlike Miro/Excalidraw, Microsoft Office/Google Workspace, or Reddit individually, it brings canvas-based creation, structured productivity blocks, and community sharing together within a single, unified application.

7. Requirement Analysis

This section analyses the requirements for the platform, covering the key features to be implemented as well as additional considerations relevant to the system's quality attributes.

7.1 Functional Requirements

- User should be able to create a workspace
- User should be able to create blocks of different document type
- Multiple users should be able to see and edit the documents together
- Users should be able to save and export projects with selective documents of their choice
- User should be able to post their project on community board
- User should be able to view, upvote, downvote and comment on public boards

7.2 Non-Functional Requirement

- **Performance:** The system should provide responsive canvas interactions and minimize delays while users draw, edit blocks, and perform other operations.
- **Real-Time Responsiveness:** Changes made by one user should be synchronized with other collaborators with minimal delay through Socket.IO.
- **Scalability:** The system should be capable of supporting increasing numbers of users, boards, and collaborative sessions without significant degradation in performance.
- **Reliability:** The system should maintain consistent board data and minimize data loss during normal operation or temporary connection failures.
- **Usability:** The interface should be simple and intuitive so that users can create boards, add content blocks, collaborate, and publish content without extensive training.
- **Security:** User authentication and authorization should protect private boards and user accounts from unauthorized access.
- **Maintainability:** The application should use a modular architecture so that individual modules such as Canvas, Collaboration, Community, Content Blocks, and Export can be developed and maintained independently.

- **Compatibility:** The web application should function correctly on commonly used modern desktop and mobile web browsers.
- **Data Integrity:** Board content, user information, comments, reactions, and version history should remain consistent and accurate when stored and retrieved.
- **Availability:** The hosted application should remain accessible to users during normal operating conditions, subject to the limitations of the selected hosting services.

8. Technology Stack

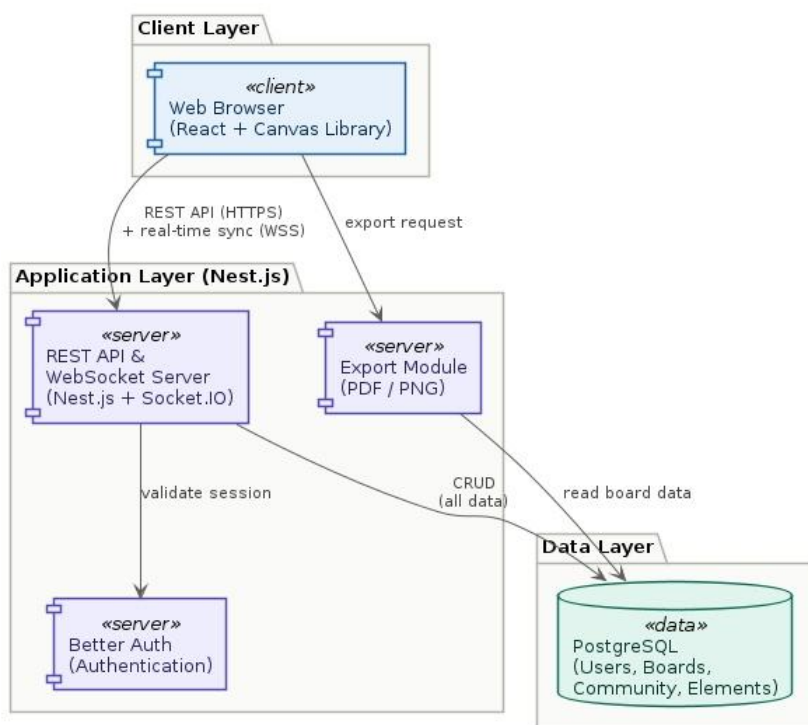
The proposed technology stack for the project is as follows:

- **Frontend:** React, along with a canvas library
- **Backend:** Nest.js with Socket.IO
- **Database:** PostgreSQL
- **Authentication:** Better Auth
- **Hosting:** name.com

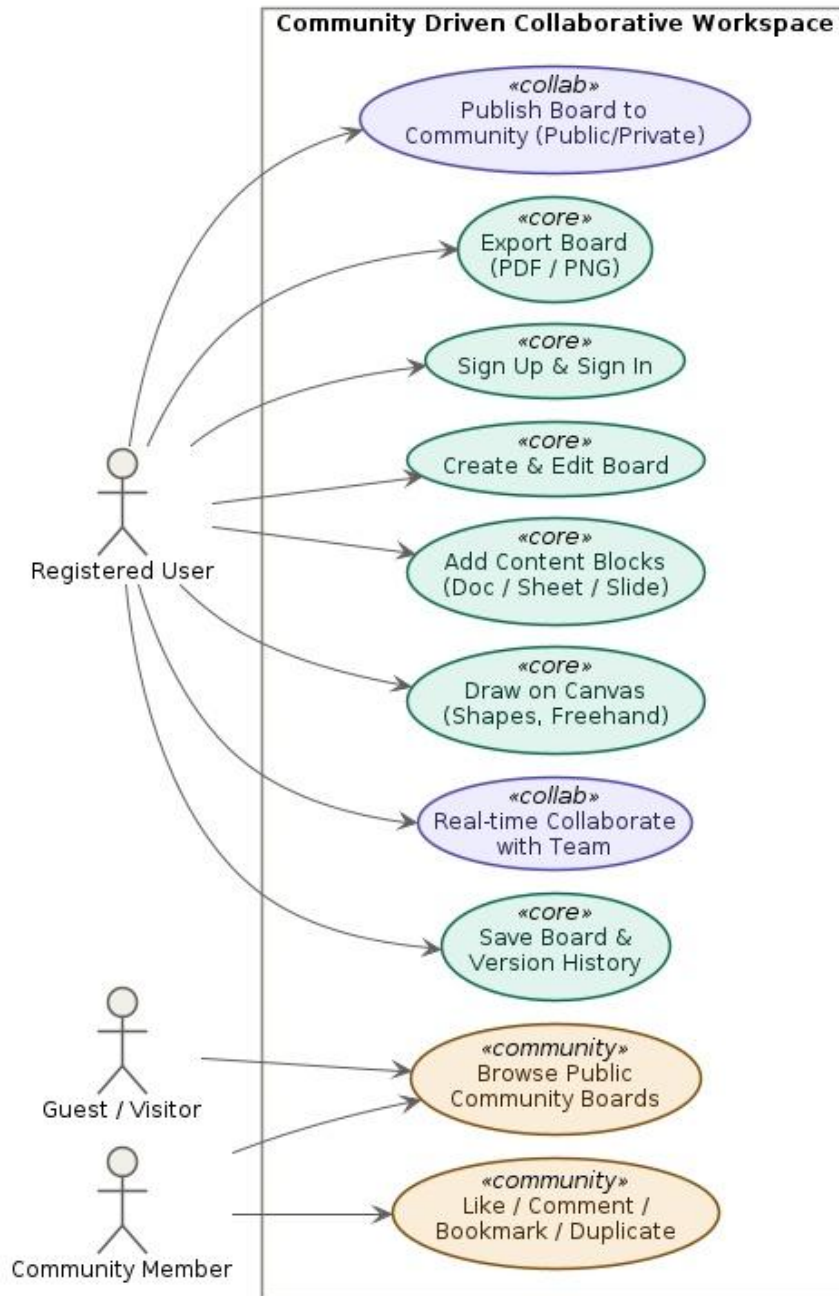
9. System Architecture

At a high level, the architecture is centered on a React frontend with a canvas library, which communicates over Socket.IO with a Nest.js backend. The backend persists board and content-block data in PostgreSQL and handles authentication through Better Auth.

The typical system workflow proceeds as follows: a user signs in and creates or opens a Board; the user adds drawings, shapes, or document/spreadsheet/slide blocks onto the infinite canvas; changes are synchronized in real time to all collaborators on the same board via Socket.IO; the user then saves the board privately or publishes it to a public or private community, where other members can view, comment on, like, and duplicate it; and finally, the user can export the board to PDF or PNG for offline use.



10. Use Case Diagrams



The primary actors and their associated use cases for the platform are summarized below:

- **Registered User:** create and edit Boards, add content blocks, collaborate with others in real time, publish Boards to the community, and export Boards.
- **Community Member:** view public Boards, and like, comment on, bookmark, and duplicate shared Boards.
- **Guest / Visitor:** browse public community Boards without editing rights.

12. Module Description

The system is organized into the following modules, each responsible for a distinct area of functionality:

- **Canvas Module:** provides the infinite canvas and the accompanying drawing tools.
- **Collaboration Module:** manages real-time multi-user editing of boards.
- **Content Blocks Module:** manages the basic document, spreadsheet, and slide blocks embedded within the canvas.
- **Community Module:** manages the publishing of boards along with community likes and comments.
- **Export Module:** manages the export of boards to PDF and PNG formats.

13. Expected Outcome

The project is expected to be both novel and feasible for the following reasons:

- It offers a unique combination of canvas-based creation, document tools, and community sharing within a single platform.
- It is built on proven, open-source technology.
- It relies on free-tier hosting, keeping development and running costs low.
- Its scope is realistic for completion by a student team within one academic year.

14. Future Enhancements

The following enhancements are planned for future iterations of the platform:

- An AI-based whiteboard assistant
- An application with offline mode
- Community events